Airline Loyalty Customer Analysis

Exploratory Data Analysis & Customer Profiling Report

Dataset: Canadian Airline Loyalty Programme
Records: 405,624 monthly flight activity records
Stack: Python | Pandas | NumPy | Matplotlib | Seaborn

1. Project Overview

This report documents the end-to-end exploratory analysis of a Canadian airline's loyalty programme. The project aimed to profile customer demographics, understand flight activity patterns, and identify the structural drivers of loyalty tier distribution — providing a foundation for data-driven decisions on programme design and customer retention.

Two source datasets were merged on a shared customer identifier:

- Customer Flight Activity — 405,624 rows of monthly flight records per customer, including flights booked, distance flown, points accumulated, and points redeemed.
- Customer Loyalty History — customer profiles including province, education level, marital status, gender, salary, loyalty card tier, enrollment type, and cancellation status.

2. Data Cleaning & Preparation

The merged dataset required three targeted cleaning interventions before analysis. Each decision was driven by the structure of the data rather than a generic imputation strategy.

2.1 Salary Nulls — Segmented Imputation

The Education column was missing the 'College' category entirely, causing all College-level customers to have null Salary values. Rather than applying the global salary mean, the imputed value was calculated as the midpoint between the two adjacent education levels — 'High School or Below' and 'Bachelor' — preserving the salary-education gradient observed in the data.

2.2 Cancellation Fields — Active Customer Encoding

Cancellation Year and Cancellation Month were null for all active customers (those who had not yet cancelled). These nulls were imputed with 0 and converted from float to int, encoding 'no cancellation has occurred' in a way that is operable and consistent with the Enrollment Year column format.

2.3 Points Accumulated — Type Correction

Points Accumulated was stored as float despite representing whole loyalty points. Values were rounded to the nearest integer and converted to int to align with Points Redeemed, which was already stored as integer. This correction removes a misleading precision signal from the data.

3. Analysis & Key Findings

Six analytical questions were investigated on the cleaned dataset. For each, the visualisation type was selected based on the structure of the variables being compared.

3.1 Seasonal Flight Activity

How are booked flights distributed across months?

Visualisation: Line plot — connects data points in chronological order to reveal temporal trends.

The data shows a clear seasonal pattern in flight bookings:

- Peak activity in July and August, corresponding to the summer holiday period.
- Lowest activity in January and February — the post-holiday trough.
- Steady growth from March through the summer peak, reflecting spring and early summer demand.

Implication: Marketing and operational planning should account for strong seasonality. The January–February trough may represent an opportunity for targeted promotions to sustain engagement during low periods.

3.2 Distance and Points Accumulation

Is there a relationship between flight distance and points accumulated?

Visualisation: Dual scatter plot — segmented by Loyalty Card tier and by Enrollment Type separately.

Key findings:

- Strong positive correlation between distance flown and points accumulated, with low dispersion. Points accumulation is directly tied to miles flown.

- Multiple distinct diagonal lines confirm that accumulation rates differ by loyalty tier — higher tiers earn points at a faster rate per mile.
- The '2018 Promotion' enrollment type shows a higher accumulation rate than standard enrollment, visible as elevated diagonal lines in the Enrollment Type scatter.

Implication: The points system is mechanistically transparent — distance drives accumulation. Tier differences and the 2018 promotion are the main sources of variation. Customers enrolled during the promotion hold a structural advantage in point accumulation.

3.3 Customer Geographic Distribution

How are customers distributed by province and country?

Visualisation: Bar plot for province distribution; pie chart for country distribution.

- Ontario, British Columbia, and Quebec are the top three provinces by customer volume, with Ontario leading by a significant margin.
- The dataset is entirely Canadian — 100% of customers are from Canada. The country-level pie chart confirms the absence of international customers in this dataset.

Implication: Loyalty programme activity is heavily concentrated in three provinces. Regional campaigns or partnerships could leverage this concentration, while also identifying lower-penetration provinces as growth opportunities.

3.4 Salary and Education Level

How does average salary compare across education levels?

Visualisation: Bar plot (mean salary by level) and box plot (full distribution).

- Clear positive gradient: higher education consistently correlates with higher average salary.
- The largest salary gap is between Master's and Doctorate level — doctoral holders earn approximately double the salary of the lowest tier.
- The 'College' category shows only a single median line in the box plot (no interquartile range), reflecting that all College values were imputed with a single calculated figure in the cleaning phase.

Implication: Education level is a reliable proxy for customer purchasing power in this dataset, which may be relevant for segmenting upgrade offers or premium product targeting.

3.5 Loyalty Card Tier Distribution

What is the proportion of customers by loyalty card type?

Visualisation: Pie chart — shows parts of a whole across a small number of categories.

- Nearly half of all customers hold the entry-level 'Star' card.
- The remaining customers are distributed across mid and upper tiers, with progressively smaller proportions at higher levels.

Implication: A large share of the customer base sits at the base of the loyalty pyramid. Understanding the barriers to progression — and whether current incentives are sufficient to drive tier advancement — is a priority question for programme design.

3.6 Customer Demographics: Marital Status and Gender

How are customers distributed by marital status and gender?

Visualisation: Count plot — compares frequency across two categorical variables simultaneously.

- Married customers are the largest segment, approximately double the size of the single customer segment.
- Gender distribution is broadly even across all marital status categories — no significant skew toward either gender within any group.

4. Summary of Findings

Area	Key Finding
Seasonality	Peak bookings July–August; trough January–February. Strong predictable cycle.
Points Accumulation	Directly proportional to distance flown. Rate varies by loyalty tier and enrollment type.
Geography	Customer base concentrated in Ontario, BC, and Quebec. 100% Canadian dataset.
Salary & Education	Clear positive gradient. Doctorate holders earn ~2x the lowest tier.
Loyalty Tiers	~50% of customers at entry-level 'Star' tier. Progression incentives may need review.
Demographics	Majority married. Gender distribution even across all segments.

5. Open Questions & Data Limitations

Several variables and relationships were identified during analysis that could not be fully explored with the available data:

- No market salary benchmarks — it is not possible to assess whether the airline's customers are over- or under-compensated relative to comparable populations, limiting the depth of salary-based segmentation.
- Cancellation drivers unknown — the dataset confirms when customers cancelled but provides no reason. A churn analysis would require additional survey or behavioural data.
- Points redeemed vs. accumulated gap — some customers accumulate significantly more points than they redeem. The reasons (programme complexity, redemption friction, lack of awareness) are not captured in the data.
- No time dimension on loyalty tier — it is unclear how long customers remain at each tier or what triggers advancement, which limits conclusions about programme effectiveness.

6. Technical Stack

Tool / Library	Usage
Python	Core language for all analysis and processing
Pandas	Data loading, merging, cleaning, aggregation
NumPy	Numerical operations and statistical calculations
Matplotlib	Base visualisation layer
Seaborn	Statistical visualisation (box plots, scatter, count plots, bar plots)
Custom src modules	EDA support functions, colour palettes, correlation helpers